Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

What is the ratio of the amount, by weight, of Metal A to Metal B in Alloy K?

1. The ratio of Metal A to Metal B in Alloy K becomes 5:6 when 40 pounds of Metal A is added to the alloy.
2. The ratio of Metal A to Metal B in Alloy K becomes 7:6 when 60 pounds of Metal A is added to the alloy.

Applied Problems

We are told how adding two different quantities of Metal A to Alloy K of Metals A and B changes the ratio between the two metals. We are asked to determine what the ratio between Metals A and B was before more of Metal A was added. Let A_1 be the quantity of metal.

1. Let A_1 be the original quantity in pounds of Metal A in the alloy before any further additions of Metal A. Let B_1 be the original quantity in pounds of Metal B in the alloy. Then, based on the information in (1), we know that $(A_1 + 40) : B_1 = 5 : 6$. Alternatively, this could be written as an equality of two fractional amounts: $(A_1 + 40)/B_1 = 5/6$. It follows from this that $5(B_1) = 6(A_1 + 40)$. This information is not sufficient alone to determine a unique value for the original ratio of Metal A to Metal B in Alloy K; NOT sufficient.
2. A similar approach to the information in (1) is applicable to the information in (2). This allows us to immediately conclude that the information in (2), like the information in (1), is also not sufficient alone to uniquely determine the original ratio of Metal A to Metal B in Alloy K; NOT sufficient.

The information gleaned from (2), as in (1), will give us an equation, derived in the same way as in (1). Then we have two equations, each with two unknowns, A_1 and B_1. Solving for A_1 and B_1 will allow us to provide a definitive answer to the question posed about the ratio of Metal A to Metal B in the original Alloy K before any further additions of Metal A. The answer you should get is the following: $A_1 : B_1 = 1 : 6$.

The correct answer is C; both statements together are sufficient.

•

Train A started from Station X toward Station Y traveling at a constant speed, and on the same day, on a parallel track, Train B started from Station Y toward Station X at a constant speed. At what time did Train A reach Station Y?

1. The two trains met at 3:00 p.m. The distance traveled by Train A was twice the distance traveled by Train B by the time they met.
2. The ratio between the speeds of Trains A and B is 3:5. Train B started from Station Y at 6:00 a.m.

Applied Problems

We are asked to determine Train A's time of arrival at Station Y.

1. This provides us with the ratio between the distance traveled by Train A to the distance traveled by Train B when the trains meet at 3:00 p.m. But we lack enough information about the speeds of the trains, neither the specific speeds nor the ratio between the speeds. Therefore, (1) is not sufficient alone to provide a definitive answer to the question posed; NOT sufficient.
2. This provides us with the ratio between the speeds of Trains A and B, and also tells us that Train B left Station Y at 6:00 a.m. But it tells us nothing about the distances that Trains A and B have traveled when they met. Therefore, (2) is not sufficient alone to provide a definitive answer to the question asked; NOT sufficient.

If the information in (1) and (2) is combined, it will be sufficient to determine the time at which Train A will reach Station Y. We have, in total, information concerning speeds, distances, and times. The distance Train A must travel from 3:00 p.m. is the distance Train B has already traveled. Train B covered that distance in 9 hours. The ratio between Train A's speed and Train B's speed is 3:5. So Train A will take 5/3 times 9 hours, i.e. 15 hours after 3:00 p.m., to arrive at Station Y. Therefore, the total information in (1) and (2) is sufficient to provide a definitive answer to the question, but neither (1) nor (2) is sufficient alone.

The correct answer is C; both statements together are sufficient.

•

Pipes A and B are the pipes that fill a tank. Pipes C and D are the pipes that empty the same tank. What is the capacity of the tank?

1. Pipe A is twice as efficient as Pipe C, that is, the volume of water per hour that Pipe A moves is twice the volume per hour that Pipe C moves. Pipe A and Pipe B are equally efficient. When Pipes C and D are closed, Pipes A

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

per hour that Pipe C moves. Pipe A and Pipe B are equally efficient. When Pipes C and D are closed, Pipes A and B together fill the empty tank in 5 hours. Pipe C empties 6,000 liters per hour.

2. Pipe B is three times as efficient as Pipe D.

Applied Problems

We are asked to determine the volume of a tank that has two pipes, A and B, that can fill it and two pipes, C and D, that can drain it. Is there sufficient information to find the volume of the tank?

1. Since Pipe A is twice as efficient as Pipe C, and Pipe C drains 6,000 liters per hour, Pipe A fills 12,000 liters per hour. Since Pipes A and B are equally efficient, Pipe B also fills 12,000 liters per hour. If Pipes C and D, the drainage pipes, are closed, then Pipes A and B fill the tank in 5 hours. This means that the total volume of the tank is 5 times 24,000 liters, i.e., 120,000 liters. Therefore, (1) is sufficient alone to provide an answer to the question asked; SUFFICIENT.
2. The information provided in (2) is irrelevant to determining the volume of the tank. We are not given a rate of filling per hour for Pipes A or B. Therefore, (2) is not sufficient alone to provide an answer to the question asked; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

Questions 320 to 362 — Difficulty: Medium

320. What is the total number of executives at Company P?

1. The number of male executives is $\frac{3}{5}$ the number of female executives.
2. There are 4 more female executives than male executives.

Algebra Simultaneous Equations

Let M be the number of male executives at Company P and let F be the number of female executives at Company P. Determine the value of $M + F$.

1. Given that $M = \frac{3}{5}F$, it is not possible to determine the value of $M + F$. For example, if $M = 3$ and $F = 5$, then $M = \frac{3}{5}F$ and $M + F = 8$. However, if $M = 6$ and $F = 10$, then $M = \frac{3}{5}F$ and $M + F = 16$; NOT sufficient.
2. Given that $F = M + 4$, it is not possible to determine the value of $M + F$. For example, if $M = 3$ and $F = 7$, then $F = M + 4$ and $M + F = 10$. However, if $M = 4$ and $F = 8$, then $F = M + 4$ and $M + F = 12$; NOT sufficient.

Taking (1) and (2) together, then $F = M + 4$ and $M = \frac{3}{5}F$, so $F = \frac{3}{5}F + 4$. Now solve for F to get $\frac{2}{5}F = 4$ and $F = 10$. Therefore, using $F = 10$ and $F = M + 4$, it follows that $M = 6$, and hence $M + F = 6 + 10 = 16$.

The correct answer is C; both statements together are sufficient.

321. Jack picked 76 apples. Of these, he sold $4y$ apples to Juanita and $3t$ apples to Sylvia. If he kept the remaining apples, how many apples did he keep? (t and y are positive integers.)

1. $y \geq 15$ and $t = 2$
2. $y = 17$

Algebra Simultaneous Equations

Determine the value of the nonnegative integer $76 - 4y - 3t$.

1. Given that $y \geq 15$ and $t = 2$, it follows that $76 - 4y - 3t = 76 - 4y - 6 = 70 - 4y$ could have the value 70 – 4(15) = 10, 70 – 4(16) = 6, or 70 – 4(17) = 2; NOT sufficient.
2. Given that $y = 17$, it follows that $76 - 4y - 3t = 76 - 68 - 3t = 8 - 3t$ could have the value 8 – 3(1) = 5 or 8 – 3(2) = 2; NOT sufficient.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

5 or $6 = 3(2) - 2$, NOT sufficient.

Taking (1) and (2) together, it follows that $76 - 4y - 3t = 76 - 4(17) - 3(2) = 2$.

The correct answer is C; both statements together are sufficient.

322. The total price of 5 pounds of regular coffee and 3 pounds of decaffeinated coffee was \$21.50. What was the price of the 5 pounds of regular coffee?

1. If the price of the 5 pounds of regular coffee had been reduced 10 percent and the price of the 3 pounds of decaffeinated coffee had been reduced 20 percent, the total price would have been \$18.45.
2. The price of the 5 pounds of regular coffee was \$3.50 more than the price of the 3 pounds of decaffeinated coffee.

Algebra Simultaneous Equations

Let $\$x$ be the price of 5 pounds of regular coffee and let $\$y$ be the price of 3 pounds of decaffeinated coffee. Then $x + y = 21.5$, or $y = 21.5 - x$. Determine the value of x.

1. Given that $0.9x + 0.8y = 18.45$, the value of x can be determined because substituting $y = 21.5 - x$ into $0.9x + 0.8y = 18.45$ gives a single equation with unknown x, and that equation can be solved for a unique value of x; SUFFICIENT.

   Alternatively (see the Tip below), $y = 21.5 - x$ and $0.9x + 0.8y = 18.45$ are equations of nonparallel lines in the coordinate plane and thus have a single point of intersection. One way to ensure that the lines are not parallel is to solve each equation for y in terms of x and note that the coefficients of x are different. In this case, $y = 21.5 - x$ gives a coefficient of x equal to -1 and $0.9x + 0.8y = 18.45$ gives a coefficient of x equal to $-\frac{0.9}{0.8}$.

2. Given that $x = 3.5 + y$, the value of x can be determined because substituting $y = 21.5 - x$ into $x = 3.5 + y$ gives a single equation with unknown x, and that equation can be solved for a unique value of x; SUFFICIENT.

   Alternatively (see the Tip below), $y = 21.5 - x$ and $x = 3.5 + y$ are equations of nonparallel lines in the coordinate plane and thus have a single point of intersection. One way to ensure that the lines are not parallel is to solve each equation for y in terms of x and note that the coefficients of x are different. In this case, $y = 21.5 - x$ gives a coefficient of x equal to -1 and $x = 3.5 + y$ gives a coefficient of x equal to 1.

Tip: Data Insights problems involving simultaneous equations usually do not have to be solved. Instead, it is typically enough to determine whether no solution exists, exactly one solution exists, or more than one solution exists, and this can often be accomplished by graphical methods.

The correct answer is D; each statement alone is sufficient.

323. A certain painting job requires a mixture of yellow, green, and white paint. If 12 quarts of paint are needed for the job, how many quarts of green paint are needed?

1. The ratio of the amount of green paint to the amount of yellow and white paint combined needs to be 1 to 3.
2. The ratio of the amount of yellow paint to the amount of green paint needs to be 3 to 2.

Algebra Ratios; Simultaneous Equations

This problem involves 3 quantities:

The number of quarts of yellow paint, represented by Y; the number of quarts of green paint, represented by G; and the number of quarts of white paint, represented by W.

It is given that $Y + G + W = 12$.

The task is to uniquely determine the value of G, which is possible only if there is enough information to eliminate Y and W from $Y + G + W = 12$, so that only G is left.

1. Using the variables introduced above, this gives $\frac{G}{Y + W} = \frac{1}{3}$ or $3G = Y + W$. Substituting $3G$ for $Y + W$ in $Y + G + W = 12$ gives $4G = 12$, which yields a unique value of G; SUFFICIENT.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

2. Using the variables introduced above, this gives $\frac{Y}{G} = \frac{3}{2}$ or $Y = \frac{3}{2}G$. Then, by substitution, $\frac{3}{2}G + G + W = 12$. Because the value of W can vary, the value of G cannot be uniquely determined; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

324. A company produces a certain toy in only 2 sizes, small or large, and in only 2 colors, red or green. If, for each size, there are equal numbers of red and green toys in a certain production lot, what fraction of the total number of green toys is large?

1. In the production lot, 400 of the small toys are green.
2. In the production lot, $\frac{2}{3}$ of the toys produced are small.

Algebra Ratios

Let x be the number of small toys of each color, and let y be the number of large toys of each color. The contingency table below gives a summary of the information given. Determine the value of $\frac{y}{x+y}$.

	small	large	total
red	x	y	$x+y$
green	x	y	$x+y$
total	$2x$	$2y$	$2x+2y$

1. Given that $x = 400$, it is clearly not possible to determine the value of $\frac{y}{x+y} = \frac{y}{400+y}$; NOT sufficient.
2. Given that $\frac{2}{3}$ of the toys are small, then, because there are $(2x+2y)$ toys altogether and $2x$ small toys, it follows that $\frac{2x}{2x+2y} = \frac{2}{3}$. Therefore, $3(2x) = 2(2x+2y)$, or $x = 2y$, and hence $\frac{y}{x+y} = \frac{y}{2y+y} = \frac{y}{3y} = \frac{1}{3}$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

325. Jones has worked at Firm X twice as many years as Green, and Green has worked at Firm X four years longer than Smith. How many years has Green worked at Firm X?

1. Jones has worked at Firm X 9 years longer than Smith.
2. Green has worked at Firm X 5 years less than Jones.

Algebra Simultaneous Equations

Let J, G, and S be the number of years Jones, Green, and Smith, respectively, have worked at Firm X. Then $J = 2G$ and $G = 4 + S$. What is the value of G?

1. Given that $J = 9 + S$, then from $J = 2G$, it follows that $2G = 9 + S$. From $G = 4 + S$, it follows that $S = G - 4$, and therefore $G - 4$ can be substituted for S in the equation $2G = 9 + S$ to obtain $2G = 9 + (G - 4)$, or $G = 5$; SUFFICIENT.
2. Given that $G = J - 5$, it follows from $J = 2G$ that $G = 2G - 5$, or $G = 5$; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

326. On June 1, Mary paid Omar $360 for rent and utilities for the month of June. Mary moved out early, and Omar refunded the money she paid for utilities, but not for rent, for the days in June after she moved out. How many dollars did Omar refund to Mary?

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

1. Mary moved out on June 24.
2. The amount Mary paid for utilities was less than $\frac{1}{5}$ the amount Mary paid for rent.

Algebra **Applied Problems; Proportions**

This problem could be solved by making a table of possible amounts Mary could have paid for rent and utilities for the month of June, such that they total $360, as shown below.

Rent	Utilities
$300	$60
$310	$50
$320	$40
$330	$30
$340	$20
$350	$10

1. Given that Mary moved out on June 24, Omar refunded the amount she paid for 6 days of utilities or $\frac{6}{30} = \frac{1}{5}$ of the amount she paid for the whole month. As seen from the table, this amount varies, so $\frac{1}{5}$ of the total amount varies. Therefore, the amount Omar refunded to Mary cannot be uniquely determined; NOT sufficient.

   Note that a third column, as shown below, could be added to the table to confirm that $\frac{1}{5}$ of the amount she paid for utilities for the whole month varies. Adding this column will aid in determining whether (1) and (2) together are sufficient. Amounts shown in **boldface** satisfy (1).

Rent	Utilities	$\frac{1}{5}$ Utilities
$300	$60	**$12**
$310	$50	**$10**
$320	$40	**$8**
$330	$30	**$6**
$340	$20	**$4**
$350	$10	**$2**

2. Given that the amount Mary paid for utilities was less than $\frac{1}{5}$ the amount she paid for rent, the amount Omar refunded to Mary cannot be uniquely determined. To verify this, another column can be added to the table to show $\frac{1}{5}$ the amount Mary paid for rent and to show the amounts she paid for utilities that are less than $\frac{1}{5}$ the amount she paid for rent. Amounts shown *underlined* in the table below satisfy (2); NOT sufficient.

Rent	Utilities	$\frac{1}{5}$ Utilities	$\frac{1}{5}$ Rent
$300	$60	**$12**	$60
$310	<u>$50</u>	**$10**	$62
$320	<u>$40</u>	**$8**	$64
$330	<u>$30</u>	**$6**	$66
$340	<u>$20</u>	**$4**	$68

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

$350	$10	**$2**	$70

Taking (1) and (2) together, observe that there are several amounts in **boldface** in the third column that satisfy (1) and corresponding amounts underlined in the second column that satisfy (2). Therefore, the amount Omar refunded to Mary cannot be uniquely determined.

The correct answer is E; both statements together are still not sufficient.

327. A collection of 36 cards consists of 4 sets of 9 cards each. The 9 cards in each set are numbered 1 through 9. If one card has been removed from the collection, what is the number on that card?

1. The units digit of the sum of the numbers on the remaining 35 cards is 6.
2. The sum of the numbers on the remaining 35 cards is 176.

Arithmetic Place Value

The sum of all the numbers on the cards in the collection is 4(1 + 2 + 3 + … + 9) = 4(45) = 180, and hence the sum of the numbers on the remaining cards is 180 minus the number on the removed card. Therefore, if the number on the removed card is 1, 2, 3, 4, 5, 6, 7, 8, or 9, then the sum of the numbers on the remaining cards is, respectively, 179, 178, 177, 176, 175, 174, 173, 172, or 171.

1. Given that the units digit of the sum of the numbers on the remaining cards is 6, the list above shows the sum of the numbers on the remaining cards must be 176, and hence the number on the removed card must be 4; SUFFICIENT.
2. Given that the sum of the numbers on the remaining cards is 176, the list above shows the number on the removed card must be 4; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

328. What is the ratio of the number of cups of flour to the number of cups of sugar required in a certain cake recipe?

1. The number of cups of flour required in the recipe is 250 percent of the number of cups of sugar required in the recipe.
2. $1\frac{1}{2}$ more cups of flour than cups of sugar are required in the recipe.

Arithmetic Ratio and Proportion

Let F and S be the number of cups of flour and sugar, respectively, required in the cake recipe. We are to determine the ratio $F : S$.

1. Given that $F = 2.5S$, it follows that $F : S$ is 2.5:1 (if $S = 1$, then $F = 2.5$), or equivalently, 5:2; SUFFICIENT.
2. Given that $F = 1.5 + S$, it is not possible to determine the ratio $F : S$. For example, if $F = 2.5$ and $S = 1$, then $F = 1.5 + S$ and $F : S$ is 2.5:1, or equivalently, 5:2. On the other hand, if $F = 10$ and $S = 8.5$, then $F = 1.5 + S$ and $F : S$ is 10:8.5, or equivalently, 20:17; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

329. A clothing manufacturer makes jackets that are wool or cotton or a combination of wool and cotton. The manufacturer has 3,000 pounds of wool and 2,000 pounds of cotton on hand. Is this enough wool and cotton to make at least 1,000 jackets?

1. Each wool jacket requires 4 pounds of wool, and no cotton.
2. Each cotton jacket requires 6 pounds of cotton, and no wool.

Arithmetic Applied Problems

1. Given that each wool jacket requires 4 pounds of wool and no cotton, then at most a total of 750 wool jackets can be made (because $\frac{3{,}000}{4} = 750$), and possibly no other jackets. Therefore, it is possible that there is enough wool and cotton to make at least 1,000 jackets (for example, if there is enough cotton to make 250 cotton jackets), and it is possible that there is not enough wool and cotton to make at least 1,000 jackets (for example, if there is less than the amount of cotton needed to make 250 cotton jackets); NOT sufficient.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

sufficient.

2. Given that each cotton jacket requires 6 pounds of cotton and no wool, then at most a total of 333 cotton jackets can be made (because rounding $\frac{2{,}000}{6}$ down to the nearest integer gives 333), and possibly no other jackets. Therefore, it is possible that there is enough wool and cotton to make at least 1,000 jackets (for example, if there is enough wool to make 667 wool jackets), and it is possible that there is not enough wool and cotton to make at least 1,000 jackets (for example, if there is less than the amount of wool needed to make 667 wool jackets); NOT sufficient.

Taking (1) and (2) together, there is enough wool and cotton to make 750 wool jackets and 333 cotton jackets for a total of 1,083 jackets.

The correct answer is C; both statements together are sufficient.

330. Are at least 10 percent of the people in Country X who are 65 years old or older employed?

1. In Country X, 11.3 percent of the population is 65 years old or older.
2. In Country X, of the population 65 years old or older, 20 percent of those with a college degree and 10 percent of those without a college degree are employed.

Arithmetic Percents

1. Given that 11.3 percent of the population is 65 years old or older, it is not possible to determine whether at least 10 percent of those 65 years old or older are employed because no information about employment is provided; NOT sufficient.
2. Given that of those 65 years and older, 20 percent of those with a college degree and 10 percent of those without a college degree are employed, it follows that the number of employed is more than the number corresponding to 10 percent of the college degree holders and 10 percent of those without a degree. Therefore, of those 65 or older, more than 10 percent are employed; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

331. A scientist recorded the number of eggs in each of 10 birds' nests. What was the standard deviation of the numbers of eggs in the 10 nests?

1. The average (arithmetic mean) number of eggs for the 10 nests was 4.
2. Each of the 10 nests contained the same number of eggs.

Arithmetic Statistics

1. Given that the average number of eggs is 4, the standard deviation of the numbers of eggs cannot be determined. For example, if each of the 10 nests had 4 eggs, then the average number of eggs would be 4, each of the squared distances of these numbers from the average would be 0, and the average of these squared distances would be 0. Therefore, the standard deviation of the numbers of eggs—the square root of the average of these squared distances—would be $\sqrt{0} = 0$. On the other hand, if there were 5 nests each containing 3 eggs and 5 nests each containing 5 eggs, then the average number of eggs would be 4, each of the squared distances of these numbers from the average would be 1, and the average of these squared distances would be 1. Therefore, the standard deviation of the numbers of eggs—the square root of the average of these squared distances—would be $\sqrt{1} = 1$; NOT sufficient.
2. Given that each of the 10 nests contains the same number of eggs, the average number of eggs is equal to the number of eggs in each of the 10 nests. Thus, each of the squared distances of these numbers from the average is 0, and hence the average of these squared distances is 0. Therefore, the standard deviation of the numbers of eggs—the square root of the average of these squared distances—is $\sqrt{0} = 0$; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

332. A paint mixture was formed by mixing exactly 3 colors of paint. By volume, the mixture was $x\%$ blue paint, $y\%$ green paint, and $z\%$ red paint. If exactly 1 gallon of blue paint and 3 gallons of red paint were used, how many gallons of green paint were used?

1. $x = y$
2. $z = 60$

Algebra First-Degree Equations

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Algebra First-Degree Equations

Letting T represent the total number of gallons of the paint in the mixture, and B, G, and R the amounts of blue paint, green paint, and red paint, respectively, in the mixture, then $B = \frac{x}{100}T$, $G = \frac{y}{100}T$, and $R = \frac{z}{100}T$. It is given that $\frac{x}{100}T = 1$ and $\frac{z}{100}T = 3$, from which it follows that $z = 3x$. Also, since there are exactly three colors of paint, $x + y + z = 100$ and $B + G + R = T$. Determine the value of G, or equivalently, the value $\frac{y}{100}T$.

1. Given that $x = y$ and $\frac{x}{100}T = 1$, it follows that $\frac{y}{100}T = 1$; SUFFICIENT.
2. Given that $z = 60$ and $\frac{z}{100}T = 3$, it follows that $T = 5$. Knowing that the total amount of paint in the mixture is 5 gallons of which 1 gallon is blue paint and 3 gallons are red paint, the amount of green paint can be determined; SUFFICIENT.

The correct answer is D; each statement alone is sufficient.

333. For a basic monthly fee of F yen (¥F), Naoko's first cell phone plan allowed him to use a maximum of 420 minutes on calls during the month. Then, for each of x additional minutes he used on calls, he was charged ¥M, making his total charge for the month ¥T, where $T = F + xM$. What is the value of F?

1. Naoko used 450 minutes on calls the first month, and the total charge for the month was ¥13,755.
2. Naoko used 400 minutes on calls the second month, and the total charge for the month was ¥13,125.

Algebra Simultaneous Equations

1. Given that Naoko used 450 minutes and was charged ¥13,125, it follows that $F + (450 - 420)M = 13{,}755$, or $F + 30M = 13{,}755$. However, the value of F cannot be determined. For example, $F = 13{,}005$ and $M = 25$ is possible, and $F = 13{,}035$ and $M = 24$ is possible; NOT sufficient.
2. Given that Naoko used 400 minutes and was charged ¥13,125, it follows that $F = 13{,}125$, since Naoko did not use more than 420 minutes; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

334. Is the sum of the prices of the 3 books that Shana bought less than $48?

1. The price of the most expensive of the 3 books that Shana bought is less than $17.
2. The price of the least expensive of the 3 books that Shana bought is exactly $3 less than the price of the second most expensive book.

Arithmetic Operations with Integers; Order

Let B_1, B_2, and B_3 be the prices, in dollars and in numerical order, of the 3 books. Thus, $B_1 \le B_2 \le B_3$. Determine whether $B_1 + B_2 + B_3 < 48$.

1. Given that $B_3 < 17$, it is possible that $B_1 + B_2 + B_3 < 48$ (choose $B_1 = B_2 = 15$ and $B_3 = 16$), and it is possible that $B_1 + B_2 + B_3 > 48$ (choose $B_1 = B_2 = 16$ and $B_3 = 16.5$); NOT sufficient.
2. Given that $B_1 = B_2 - 3$, it is possible that $B_1 + B_2 + B_3 < 48$ (choose $B_1 = 3$, $B_2 = 6$, and $B_3 = 10$), and it is possible that $B_1 + B_2 + B_3 > 48$ (choose $B_1 = 3$, $B_2 = 6$, and $B_3 = 48$); NOT sufficient.

Taking (1) and (2) together, from (1) and the information given, it follows that $B_3 < 17$ and $B_2 \le B_3 < 17$, or $B_2 < 17$. Also, from $B_2 < 17$ and (2), it follows that $B_1 < 17 - 3$, or $B_1 < 14$. Therefore, $B_1 + B_2 + B_3$ is the sum of 3 numbers—a number less than 14, a number less than 17, and a number less than 17—and thus $B_1 + B_2 + B_3 < 14 + 17 + 17 = 48$.

The correct answer is C; both statements together are sufficient.

335. If Elena spent a total of $720 on newspapers, magazines, and books last year, what amount did she spend on newspapers?

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

newspapers?

1. Last year, the amount that Elena spent on magazines was 80 percent of the amount that she spent on books.
2. Last year, the amount that Elena spent on newspapers was 60 percent of the total amount that she spent on magazines and books.

Arithmetic Statistics

Let $\$N$, $\$M$, and $\$B$ be the amount spent, respectively, on newspapers, magazines, and books. We are given that $N + M + B = 720$ and are to determine the value of N.

1. Given that $M = 0.8B$, it follows from $N + M + B = 720$ that $N + 1.8B = 720$. By considering different values of B (for example, $B = 10$ and $B = 100$), we can show that more than one value of N is possible, and therefore it is not possible to determine the value of N. For example, if $B = 10$, then $M = 0.8B = 8$ and $N = 720 - 1.8B = 702$. On the other hand, if $B = 100$, then $M = 0.8B = 80$ and $N = 720 - 1.8B = 540$; NOT sufficient.
2. Given that $N = 0.6(M + B)$, it follows from $N + M + B = 720$ that $0.6(M + B) + (M + B) = 720$, or $1.6(M + B) = 720$. Therefore, $M + B = 720 \div 1.6 = 450$, and hence $N = 0.6(M + B) = 0.6 \times 450 = 270$; SUFFICIENT.

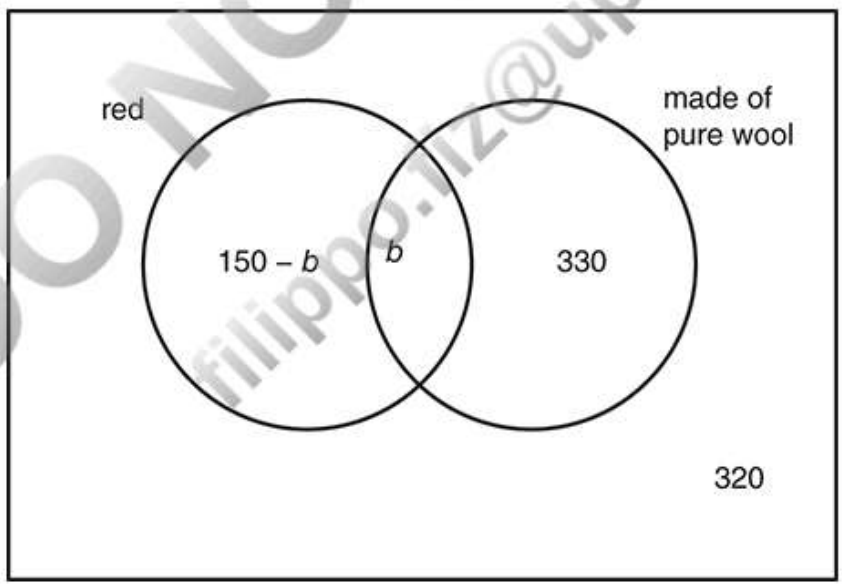
The correct answer is B; statement 2 alone is sufficient.

336. Of the 800 sweaters at a certain store, 150 are red. How many of the red sweaters at the store are made of pure wool?

1. 320 of the sweaters at the store are neither red nor made of pure wool.
2. 100 of the red sweaters at the store are not made of pure wool.

Arithmetic Statistics

1. Given that 320 of the sweaters are neither red nor made of pure wool, it follows that the 800 – 320 = 480 remaining sweaters are red, made of pure wool, or both. Since 150 of the sweaters are red, 480 − 150 = 330 of the sweaters are not red and are made of pure wool. However, the number of red sweaters made of pure wool cannot be determined. Letting b be this number, the Venn diagram below shows that any integer between 0 and 150, inclusive, can be a possible value of b; NOT sufficient.



2. Given that 100 of the red sweaters are not made of pure wool, it follows that the 150 – 100 = 50 remaining red sweaters are made of pure wool; SUFFICIENT.

The correct answer is B; statement 2 alone is sufficient.

337. A certain computer company produces two different monitors, P and Q. In 2010, what was the net profit from the sale of the two monitors?

1. Of the company's expenses in 2010, rent and utilities totaled $500,000.
2. In 2010, the company sold 50,000 units of Monitor P at $300 per unit and 30,000 units of Monitor Q at

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

2. In 2010, the company sold 50,000 units of Monitor P at $300 per unit and 30,000 units of Monitor Q at $650 per unit.

Arithmetic Applied Problems

The net profit from the sale of the two monitors is the total revenue, $\$R$, from the sale of the two monitors minus the total cost, $\$C$, of the two monitors, or $\$R - \$C$. Determine the value of $R - C$.

1. Given that the rent and utilities totaled $500,000, it is not possible to determine the value of $R - C$ because we only know that $C \geq 500{,}000$, and we do not know anything about the value of R; NOT sufficient.
2. Given that $\$R = 50{,}000(\$300) + 30{,}000(\$650)$, it is not possible to determine the value of $R - C$ because we do not know anything about the value of C; NOT sufficient.

Taking (1) and (2) together, it is not possible to determine the value of $R - C$ because from (2) the value of R is fixed, and from both (1) and (2) we can only conclude that $C \geq 500{,}000$, and therefore more than one value of $R - C$ is possible.

The correct answer is E; both statements together are still not sufficient.

338. A conveyor belt moves bottles at a constant speed of 120 centimeters per second. If the conveyor belt moves a bottle from a loading dock to an unloading dock, is the distance that the conveyor belt moves the bottle less than 90 meters? (1 meter = 100 centimeters)

1. It takes the conveyor belt less than 1.2 minutes to move the bottle from the loading dock to the unloading dock.
2. It takes the conveyor belt more than 1.1 minutes to move the bottle from the loading dock to the unloading dock.

Arithmetic Applied Problems

Since the rate at which the conveyor belt moves is given as 120 centimeters per second, which is equivalent to $\frac{120}{100} = 1.2$ meters per second, the conveyor will move less than 90 meters if it moves for less than $\frac{90}{1.2} = 75$ seconds.

1. This indicates that the length of time the conveyor belt moves is less than 1.2 minutes, which is equivalent to (1.2)(60) = 72 seconds and 72 < 75; SUFFICIENT.
2. This indicates that the length of time the conveyor belt moves is more than 1.1 minutes, which is equivalent to (1.1)(60) = 66 seconds, but does not indicate how much more than 66 seconds the conveyor moves. If the conveyor moves for 70 seconds (70 > 66), for example, it moves less than 90 meters, since 70 < 75. However, if the conveyor belt moves for 80 seconds (80 > 66), then it moves more than 90 meters, since 80 > 75; NOT sufficient.

The correct answer is A; statement 1 alone is sufficient.

339. In a product test of a common cold remedy, x percent of the patients tested experienced side effects from the use of the drug and y percent experienced relief of cold symptoms. What percent of the patients tested experienced both side effects and relief of cold symptoms?

1. Of the 1,000 patients tested, 15 percent experienced neither side effects nor relief of cold symptoms.
2. Of the patients tested, 30 percent experienced relief of cold symptoms without side effects.

Algebra Sets

The Venn diagram on the following page represents the numbers of patients who experienced neither, one, or both side effects and relief. In the diagram,

$a = \text{number}$ who experienced side effects only,

$b = \text{number}$ who experienced both,

$c = \text{number}$ who experienced relief only,

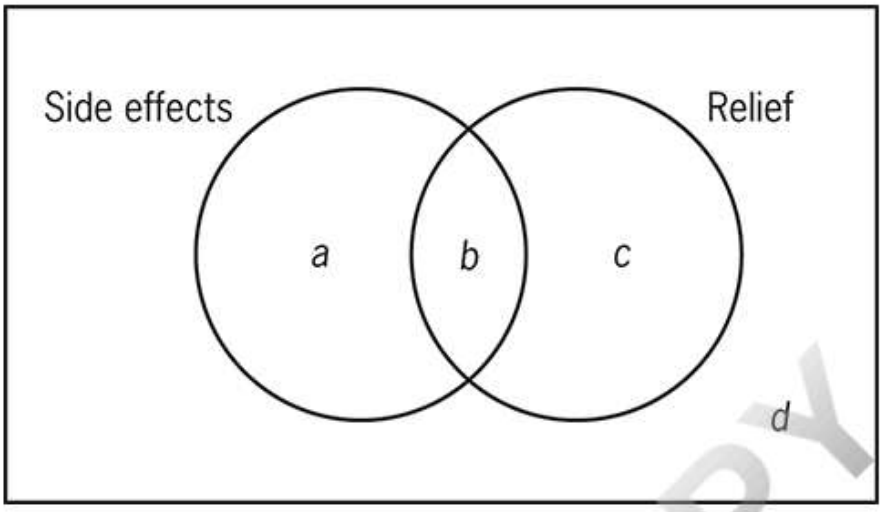
$d = \text{number}$ who experienced neither,

where $T = a + b + c + d$ is the total number of patients. Note that $a + b = \frac{x}{}T$ and $b + c = \frac{y}{}T$.

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

where $T = a + b + c + d$ is the total number of patients. Note that $a + b = \frac{\ }{100}T$ and $b + c = \frac{\ }{100}T$. Determine the value of $\frac{b}{T}$.



1. Given that 1,000 patients were tested and (0.15)(1,000) = 150 patients experienced neither side effects nor relief, it follows that $a + b + c + d = 1{,}000$ and $d = 150$. Therefore, $a + b + c + 150 = 1{,}000$, or $a + b + c = 850$. However, there is no information about how the remaining 850 patients are assigned to the other categories. Thus, the percent of patients who experienced both side effects and relief could be any value between 0% and 85%; NOT sufficient.
2. Given that 30% of the patients experienced relief without side effects, there is no information about how the remaining 70% of the patients are assigned to the other categories. Thus, the percent of patients who experienced both side effects and relief could be any value between 0% and 70%; NOT sufficient.

Taking (1) and (2) together, it follows that 150 patients experienced neither side effects nor relief and (0.3)(1,000) = 300 patients experienced relief only. Therefore, from $a + b + c + d = T$ it follows that $a + b + 300 + 150 = 1{,}000$, or $a + b = 550$. However, it is still not possible to determine a unique value of $\frac{b}{T} = \frac{b}{1{,}000}$, the percent of patients who experienced both side effects and relief. For example, it is possible that $a = 300$ and $b = 250$ (that is, the percent who experienced both could be 25%), and it is possible that $a = 250$ and $b = 300$ (that is, the percent who experienced both could be 30%).

The correct answer is E; both statements together are still not sufficient.

340. Three roommates—Bela, Gyorgy, and Janos—together saved money for a trip. The amount that Bela saved was equal to 8% of his monthly income. The amount that Gyorgy saved was exactly $\frac{1}{3}$ of the total amount saved by all 3 roommates. What was the total amount saved for the trip by all 3 roommates?

1. Bela had a monthly income of $2,000.
2. Janos saved 1.5 times as much for the trip as Bela.

Algebra First-Degree Equations

Letting B, G, and J represent the amounts saved by Bela, Gyorgy, and Janos, respectively, then $B = 0.08I$, where I represents Bela's monthly income, $G = \frac{1}{3}(B + G + J)$, and $G = \frac{1}{2}(B + J)$. Determine the value of $B + G + J$.

1. This allows the value of B to be determined but gives no information about the values of G or J; NOT sufficient.
2.

DO NOT COPY
filippo.fiz@uptoten.it